

ADVANCED REVIEW



WILEY

Explaining artificial intelligence with visual analytics in healthcare

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Funding information

Fonds Wetenschappelijk Onderzoek, Grant/Award Number: G0A3319N; Javna Agencija za Raziskovalno Dejavnost RS, Grant/Award Number: ARRS-N2-0101

Edited by: Justin Wang, Associate Editor and Witold Pedrycz, Editor-in-chief

Abstract

To make predictions and explore large datasets, healthcare is increasingly applying advanced algorithms of artificial intelligence. However, to make well-considered and trustworthy decisions, healthcare professionals require ways to gain insights in these algorithms' outputs. One approach is visual analytics, which integrates humans in decision-making through visualizations that facilitate interaction with algorithms. Although many visual analytics systems have been developed for healthcare, a clear overview of their explanation techniques is lacking. Therefore, we review 71 visual analytics systems for healthcare, and analyze how they explain advanced algorithms through visualization, interaction, shepherding, and direct explanation. Based on our analysis, we outline research opportunities and challenges to further guide the exciting rapprochement of visual analytics and healthcare.

This article is categorized under:

Application Areas > Health Care

Fundamental Concepts of Data and Knowledge > Explainable AI

Technologies > Visualization

KEYWORDS

healthcare, visual analytics, XAI

1 | INTRODUCTION

Healthcare professionals are increasingly acquiring vast amounts of electronic health records, analyzing these data with advanced algorithms like artificial intelligence (AI), and basing decisions on the algorithmic outcomes (Miotto et al., 2018). Countless examples illustrate the rise of AI in healthcare: Stiglic et al. (2018) and Kopitar et al. (2020) built predictive models for chronic diseases, X. Liu et al. (2019) detected diseases from medical imaging with deep learning, Viani et al. (2021) and Carriere et al. (2021) applied natural language processing to extract disease onset from textual health records and to assist with rehabilitation assessment and treatment, and so on.

The shift towards “big data” and AI comes with tremendous opportunities for healthcare, but also entails important challenges (Ahmad et al., 2018). A prominent challenge is that well-performing techniques such as deep learning generally yield “black box” models: understanding how they establish outputs is hard or even infeasible. Many healthcare stakeholders deem it unacceptable to fully rely on “black boxes,” and call for explaining algorithmic decision processes. This call is further reinforced by medico-legal and ethical requirements, and regulations on AI use like the European GDPR, which endorses a right to explanation (Goodman & Flaxman, 2017).

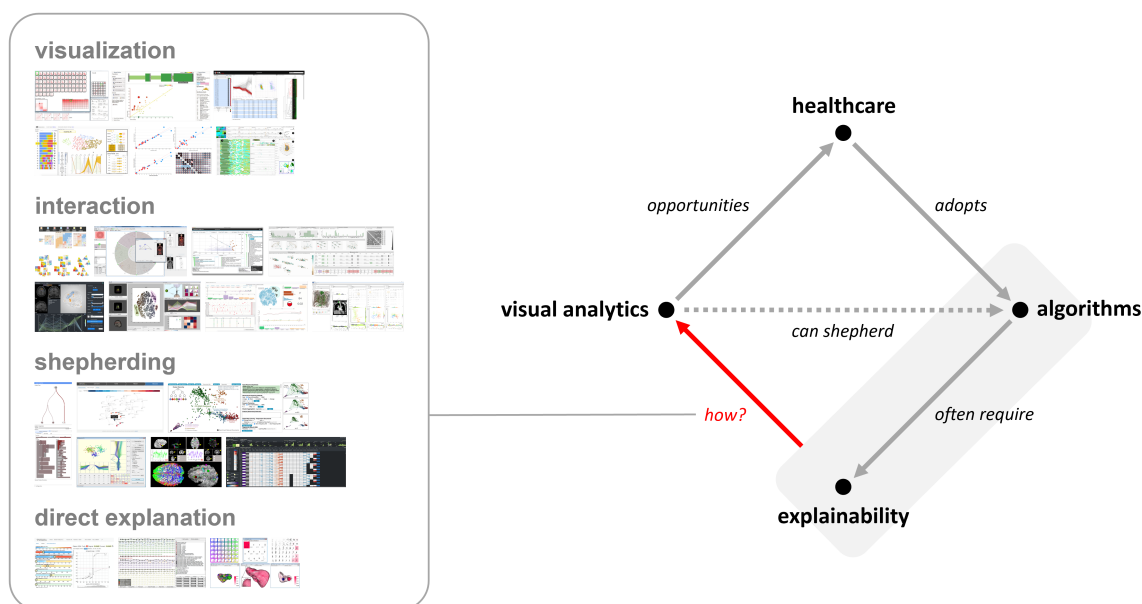


FIGURE 1 Healthcare increasingly adopts advanced algorithms, and often requires explanations for the algorithmic process. Visual analytics can provide insights in algorithms through visualization, interaction, shepherding and direct explanations. Thus, visual analytics holds important opportunities for healthcare.

Constructing explanations for AI models is the holy grail in *explainable artificial intelligence* (XAI), a melting pot of research fields like cognitive psychology, human–computer interaction, and computer science (Hind, 2019). A promising approach for XAI is *visual analytics*. This subfield of information visualization fosters analytical reasoning through interactive visual interfaces (Cui, 2019; Ham, 2010; Keim et al., 2008): by visually exploring data and iteratively refining hypotheses, users can discover complex relations in large datasets, detect biases, and get insights in how algorithms work (e.g., through *shepherding*, which is controlling the algorithmic process).

Many visual analytics systems have been developed for healthcare, but a clear overview of their explanation techniques is lacking. Therefore, we review visual analytics systems that incorporate advanced algorithms, and that were either specifically designed for a healthcare context, or evaluated therein (see Figure 1). Our contribution is twofold. First, we showcase the potential of visual analytics for explaining algorithms according to four perspectives, enclosed in our research questions:

- **RQ1.** How do visual analytics systems visualize the outcomes of advanced algorithms?
- **RQ2.** Which interactions do visual analytics systems support?
- **RQ3.** How do visual analytics systems support shepherding of advanced algorithms?
- **RQ4.** How can visual analytics systems explain advanced algorithms directly?

Second, we analyze main trends, opportunities, and remaining challenges for visual analytics in healthcare. Along the way, we report which advanced algorithms are incorporated in visual analytics systems for healthcare, and for which purposes they are used. We present our findings with an interdisciplinary audience in mind, and thus hope to further strengthen the bridge between visual analytics, AI, and healthcare.

2 | BACKGROUND AND RELATED WORK

Our review touches upon healthcare, advanced algorithms, explainable AI, and visual analytics. This section presents relevant work in the intersection of these domains.

2.1 | Explainable artificial intelligence

XAI encompasses a huge collection of intertwined topics, including trust, fairness, bias, causality, accountability, privacy, and reasoning (Abdul et al., 2018). One side-effect of this rich mix is that researchers have not yet agreed upon a rigorous definition for explainability, and often interchange it with interpretability, understandability, or intelligibility (Gilpin et al., 2018; Lipton, 2018).

Human-computer interaction recognizes that the meaning of explainability and its requirements depend on the target user and application context. Mohseni et al. (2020) classified target users into AI novices, data experts, and AI experts, each needing unique design goals and evaluation measures. Wang et al. (2020) and Ahmad et al. (2018) pointed out that the importance of explanations depends on the healthcare application: they are crucial when care is affected, but less pressing for treatment cost prediction. To determine a suitable explanation level, Vellido (2020) argued to integrate healthcare experts in the design of data analysis interpretation strategies.

Even though explainability lacks a formal definition, the AI community has developed many explanation techniques for AI models (Adadi & Berrada, 2018). Guidotti et al. (2019) categorized these techniques according to how they open the “black box” problem: by explaining the model itself, by explaining the outcomes, or by inspecting the model. Stiglic et al. (2018) and Du et al. (2019) categorized explanation techniques by scale (local vs. global) and type (model-specific vs. model-agnostic): local explanations focus on a single instance, whereas global explanations try to explain the entire model; model-specific explanations are only applicable for particular models (e.g., deep learning models, Montavon et al., 2018), whereas model-agnostic explanations can explain any model (Ribeiro et al., 2016).

2.2 | Visual analytics for explainable artificial intelligence

Many authors have surveyed visual analytics systems in the scope of AI. Some surveys mainly focus on the machine learning aspect. For example, S. Liu et al. (2017) classified visual analytics systems by whether they are intended to understand, diagnose or refine machine learning models; Endert et al. (2017) considered the machine learning type and the interaction intent. Other surveys rather focus on the visualization aspect. For example, Lu et al. (2017) categorized predictive visual analytics systems based on their interaction methods and prediction tasks; Hohman et al. (2019) discussed the why, who, what, how, when, and where of visualizing deep learning models. Chatzimpampas et al. (2020) covered both machine learning and visualization aspects in a review on enhancing trust with interactive visualizations: their fine-grained classification covers interaction type, machine learning model, and trust level.

2.3 | Visual analytics in healthcare

Ever since visual analytics emerged, healthcare has been recognized as one of its most promising application areas (Keim et al., 2008; Thomas & Kielman, 2009) because of the many opportunities for clinicians, patients, researchers, and other healthcare stakeholders (Caban & Gotz, 2015). The fruitful interplay between healthcare, medical visualization, and visual analytics produced an extensive jargon, including visual intelligent decision support systems (Ltifi & Ayed, 2016), clinical informatics (Simpao et al., 2014), and health informatics (Wu et al., 2019).

Despite the diverging terminology, a lot of interesting work has been presented in different healthcare areas, for example, population health services (Chishtie et al., 2020), prevention of disease outbreaks (Preim & Lawonn, 2020), and cancer-related genomics (Qu et al., 2019). In biomedics, Sturm et al. (2015) categorized existing work on interactivity level versus analysis type and visualization technique, and Turkay et al. (2014) classified visual analysis tools by their analytical task and integration of computational methods. Finally, Rostamzadeh et al. (2020) and West et al. (2015) reviewed interactive visualizations of electronic health records, and Wang et al. (2011) presented case studies and design guidelines based on their experiences with *Lifelines2*.

To conclude, a rich set of surveys highlights the importance of explaining machine learning. However, the general surveys on visual analytics in Section 2.2 are not framed in a healthcare context and do not discuss its specific requirements. In contrast, the reviews in Section 2.3 are healthcare-oriented, but do not focus on explanations or only cover a specific healthcare subdomain. To shed light on the intersection of healthcare, visual analytics and explanation

techniques, we review visual analytics systems for healthcare that facilitate algorithmic explainability. We also extend the scope from machine learning to advanced algorithms in general.

3 | PAPER COLLECTION AND CLASSIFICATION PROCESS

Starting from the key reviews in Section 2, we iteratively compiled the search query in Table 1 to target interactive visualizations that were specifically developed for or applied in a healthcare context, and that involve at least one advanced algorithm. Our query also considers diverging terminology for similar concepts: for example, “decision support systems” in healthcare can fit with what the visualization community considers as interactive visual dashboards.

In January 2021, the first author queried Scopus, and then screened 1908 matches based on their abstract (285 remaining) and full-text (83 remaining, including 12 overview papers). We excluded papers that present fully static or interaction-limited visualizations (i.e., first level of integration in Turkay et al., 2014), solely discuss image processing outcomes, do not allow for data analysis, present statistical or visualization software, or do not involve advanced algorithms. The latter condition excluded dashboards like *LifeLines2* (Wang et al., 2011). Finally, the first author classified all included papers in Tables 2 and 3, inspired by existing frameworks for activity types (Rostamzadeh et al., 2020), algorithmic classes (Endert et al., 2017), interaction types (Yi et al., 2007), and explanation techniques (Guidotti et al., 2019; Mohseni et al., 2020).

Sections 4 to 7 present Tables 2 and 3 in detail. Each section treats one technique to obtain insights in advanced algorithms: visualization, interaction, shepherding, or direct explanation.

4 | VISUALIZING ALGORITHMIC OUTCOMES IN VISUAL ANALYTICS

A first way to gain insights in advanced algorithms is to visualize their outcomes. Based on the first two column groups in Table 2, we discuss several visualization approaches (cfr. RQ1) for distinct algorithmic families, and uncover the healthcare activities for which visual analytics systems in our sample are used.

4.1 | Activity

Rostamzadeh et al. (2020) divided healthcare activities into interpreting, predicting, and monitoring. ■ *Interpretation* is the most frequently supported activity in our sample (55/71). Thus, most visual analytics systems are geared toward exposing patterns in data, and discovering relations among features. Less common is ■ *Prediction* whereby outcomes are anticipated or hypotheses are formed based on the available data (16/71). Finally, ■ *Monitoring* is absent in our sample: no visual analytics systems help to manage recurrent or chronic diseases.

4.2 | Algorithm

The visual analytics systems in our sample incorporate a multitude of algorithmic families. *Clustering/similarity* is the most common family (38/71): its algorithms are mainly used for interpreting, which explains that activity's prevalence.

TABLE 1 Query used for paper selection (categories were combined with an AND operator)

Category	Keywords
Algorithm	ai OR algorithm* OR artificial intelligence OR automated OR big data OR data mining OR deep learning OR machine learning OR predict*
Analytics	analytics OR data analy* OR decision support OR electronic health records
Healthcare	*medic* OR bioinformatics OR clinic* OR health*
Interaction	explor* OR interact*
Visualization	dashboard OR graphic* OR interface OR visual*

TABLE 2 Classification of 71 visual analytics systems in our sample according to healthcare activity types, present algorithms, and interaction types

	ACTIVITY		ALGORITHM										INTERACTION							
	Interpretation	Prediction	Anomaly detection	Artificial neural network	Classical statistics	Classification	Clustering/similarity	Data mining	Dimension reduction	Feature selection	Segmentation	Other	Abstract/elaborate	Connect	Encode	Explore	Filter	Reconfigure	Select	Shepherd
Abbasloo et al. (2019)	•			•										•	•	•	•	•	•	•
Abdullah et al. (2020)	•						•			•				•	•					•
Afzal et al. (2011)		•										•				•				•
Alsaad et al. (2019)		•	•			•														•
Barlowe et al. (2013)	•						•							•				•	•	•
Behrisch et al. (2018)	•											•				•				
Borland et al. (2020)	•												•							•
Brunker et al. (2019)	•			•										•						•
Cao et al. (2011)	•												•	•					•	•
Clark et al. (2017)	•				•															•
Dang et al. (2015)	•						•						•					•	•	•
Dingen et al. (2019)		•		•															•	•
Dixit et al. (2017)	•					•			•						•				•	•
Fang et al. (2017)	•									•										•
Farag et al. (2015)	•													•						•
Feller et al. (2018)	•						•			•										•
Geurts et al. (2015)	•											•								•
Gotz et al. (2011)	•						•						•							•
Gotz et al. (2014)	•																			•
Gotz et al. (2020)	•		•				•							•				•		•
Guo et al. (2020)	•			•						•			•							•
Guo et al. (2018)	•						•						•					•		•
Herold et al. (2010)	•						•													•
Hinterberg et al. (2015)	•			•																•
Huang et al. (2015)	•						•											•		•
Huang et al. (2019)	•	•					•						•							•
Hund et al. (2016)	•																			•
Hur et al. (2020)	•	•		•					•											•
Ji et al. (2017)	•												•							•
Ji et al. (2019a)	•						•						•							•
Ji et al. (2019b)	•			•			•						•							•
Jönsson et al. (2019)	•			•											•					•
Kakar et al. (2019)	•								•											•
Klemm et al. (2014)	•						•						•							•
Klimov et al. (2015)	•									•										•
Kovalerchuk et al. (2012)	•									•										•
Krause et al. (2014)	•					•							•			•				•
Krause et al. (2016)	•						•						•							•
Krause et al. (2018)	•						•						•							•
Kumar et al. (2015)	•			•			•				•		•							•
Kwon et al. (2018)	•			•			•						•			•				•
Kwon et al. (2019)	•	•		•						•			•			•				•
Kwon et al. (2020)	•								•			•	•							•
L'Yi et al. (2015)	•						•						•							•
L'Yi et al. (2017)	•	•		•									•							•
Lamy and Tsopra (2019)	•	•		•									•							•
Li et al. (2012)	•			•			•			•										•
Li et al. (2020)	•	•		•									•							•
Liao et al. (2017)	•		•				•			•			•			•				•
Males et al. (2020)	•			•									•							•
Malik et al. (2015)	•			•									•							•
Moschonas et al. (2016)	•												•							•
Müller et al. (2020)	•						•					•	•							•
Nauta et al. (2020)	•	•		•			•			•			•							•
Nguyen et al. (2011)	•						•				•		•			•				•
Nguyen et al. (2012)	•			•			•						•							•
Raidou et al. (2016a)	•			•			•			•			•							•
Raidou et al. (2016b)	•									•		•	•							•
Riegler et al. (2016)	•						•						•				•			•
Santamaría et al. (2008)	•						•						•							•
Santamaría et al. (2019)	•								•				•			•				•
Seo and Shneiderman (2002)	•						•						•							•
Song et al. (2017)	•			•									•			•				•
Spitz et al. (2020)	•	•				•							•							•
Stolper et al. (2014)	•								•				•			•				•
Verma et al. (2017)	•	•		•			•						•					•	•	•
Von Landesberger et al. (2013)	•						•			•		•	•							•
Widanagamaachchi et al. (2017)	•			•			•						•							•
Xing et al. (2014)	•			•			•						•		•					•
Yu et al. (2017)	•		•	•			•			•			•							•
Zhao et al. (2017)	•								•				•							•
Total																				
	55	16	2	8	20	11	38	11	18	3	4	5	51	38	20	19	41	38	58	32

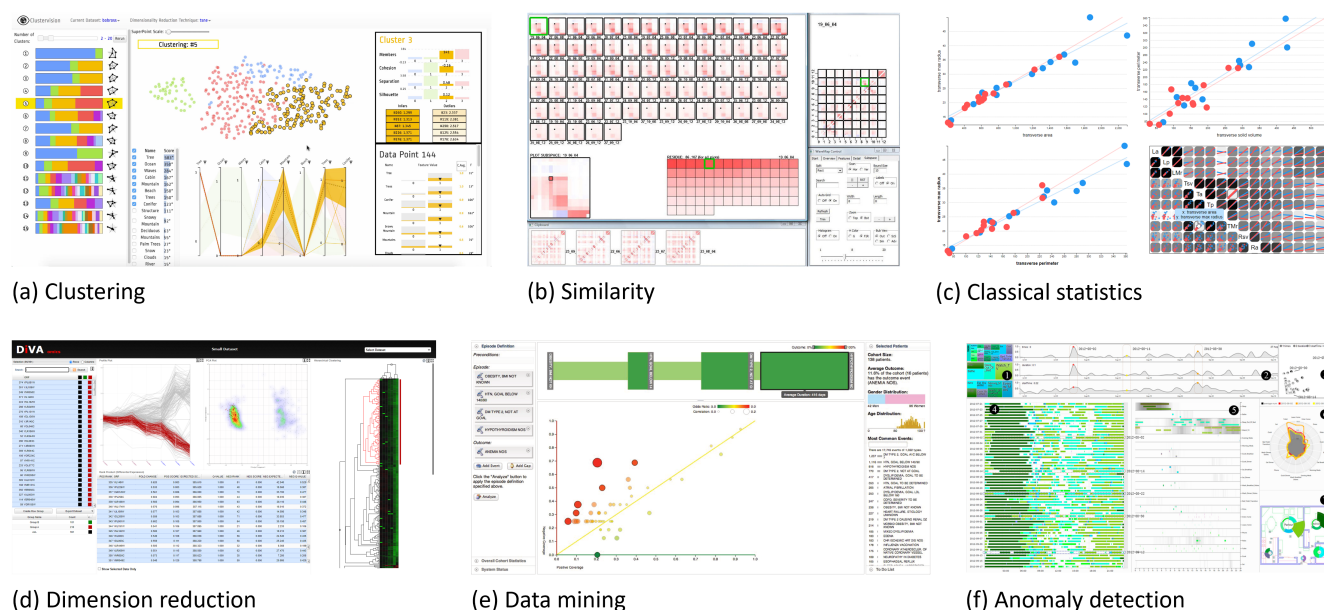


FIGURE 2 Examples of visualizing algorithmic outcomes. (a) Clustering outcomes in a scatter plot and parallel coordinates (Kwon et al., 2018); (b) plots ranked according to similarity to the first one (Barlowe et al., 2013); (c) classical regression lines in scatter plots (Males et al., 2020); (d) top: Parallel coordinates and projection plot with heat map after dimension reduction. Right: Heat map matrix and dendrogram of hierarchical clustering (Fargat et al., 2015); (e) results of sequential pattern mining in a scatter plot (Gotz et al., 2014); (f) anomalous activities highlighted in several visualizations (Liao et al., 2017)

■ **Clustering** algorithms group similar data points, say similarly nutritious meals (Feller et al., 2018) or genomes (Seo & Shneiderman, 2002). Results from k-means and k-nearest neighbors are usually visualized in scatter plots with dots colored according to their cluster (e.g., Klemm et al., 2014; X. Ji, Shen, et al., 2019, Figure 2a), or in projection plots after dimension reduction (e.g., R. Guo et al., 2020; S.-Y. Ji et al., 2017). To avoid dimensionality reduction, Abdullah et al. (2020) and Kwon et al. (2018; Figure 2a) use parallel sets: axes that represent features are connected by ribbons to show the proportional distribution of feature combinations for each cluster. Alternatively, Gotz et al. (2011) and Cao et al. (2011; Figure 4a) visualize clusters of patients as Voronoi treemaps, where cells represent features of patients. Interestingly, L'Yi et al. (2015) apply parallel sets to compare clustering algorithms. Two more clustering algorithms are biclustering and hierarchical clustering. Santamaria et al. (2008) visualize biclustered microarray data as a Venn diagram. Cluster hierarchies are typically visualized as dendrograms next to a heat map matrix (e.g., Yu et al., 2017; Farag et al., 2015; Figure 2d). Other visualizations are: an expandable list of cluster representatives (Raidou, Marcelis, et al., 2016), inductively grouped circles (Behrisch et al., 2018), and a time line of clusters at a given hierarchy level (Widanagamaachchi et al., 2017).

We classified ■ **Similarity** measures like cosine, Jaccard and Hellinger distance together with clustering, because they are often used to group data. For example, Borland et al. (2020; Figure 5d) hierarchically aggregate similar events in an icicle plot to track selection bias in patient cohorts; Barlowe et al. (2013; Figure 2b) rank protein flexibility plots by similarity to a target.

Next comes ■ **Classical statistics** (20/71), which illustrates that current systems still heavily rely on non-machine learning algorithms. *Classical statistics* includes three techniques. (1) Correlation analysis: cells in correlation matrices are typically color-coded to reveal patterns and outliers (e.g., Males et al., 2020; Song et al., 2017). (2) (Non)parametric hypothesis testing: Abbasloo et al. (2019) and Jönsson et al. (2019; Figure 3b) highlight brain regions that significantly differ for people with and without a condition; Malik et al. (2015) visualize patients that survived and died after certain event sequences as back-to-back bar charts, and indicate significant differences. (3) Regression: Males et al. (2020; Figure 2c) compare two groups' colon morphology in overlaid scatter plots and regression lines; Verma et al. (2017) predict adverse drug reactions with logistic regression, and visualize the confidence as ribbon width in a Sankey diagram.

The following techniques are more commonly associated with AI: *Dimension reduction* (18/71), *Data mining* and *Classification* (both 11/71). ■ **Dimension reduction** projects multidimensional data to 2D or 3D with principal

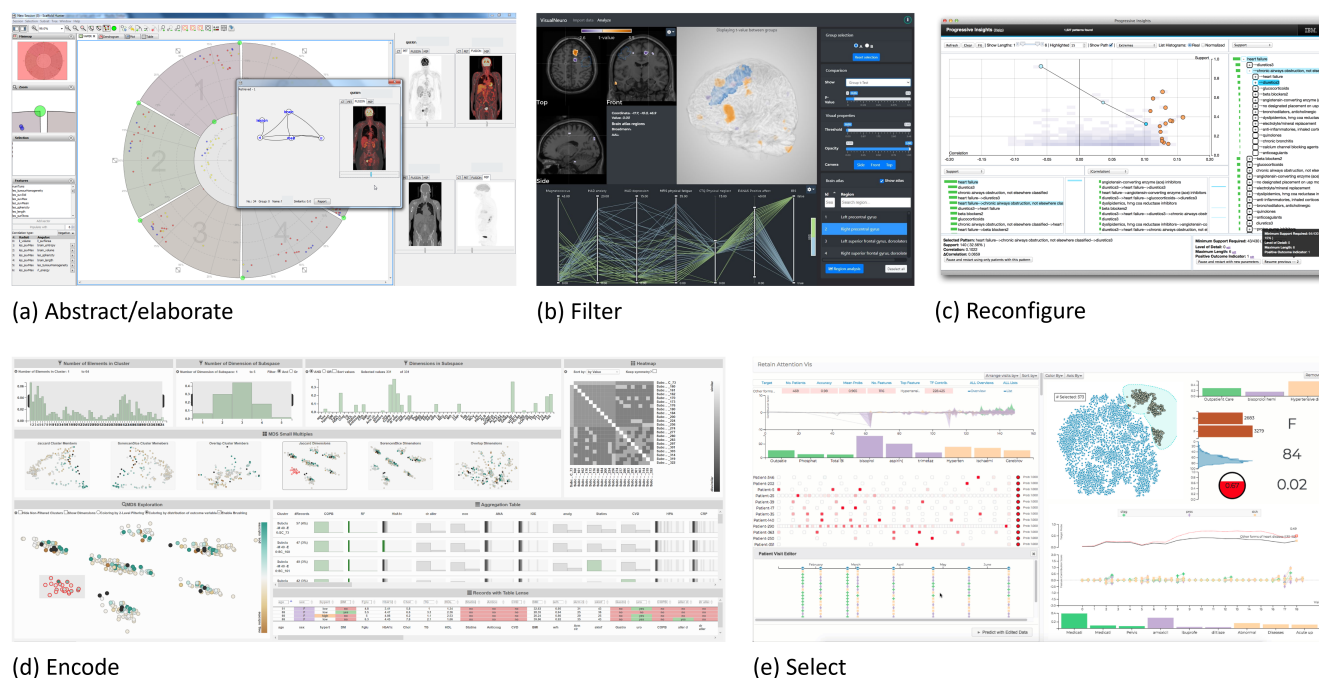


FIGURE 3 Examples of interaction with visualizations. (a) Abstract or elaborate information with zooming (Kumar et al., 2015); (b) filter data with sliders in parallel coordinates (Jönsson et al., 2019); (c) reconfigure a scatter plot by changing axes' features, and connect a selected data point with related points (Stolper et al., 2014); (d) encode data differently by adding spikes to points in a scatter plot, and brush-and-link (Hund et al., 2016); (e) select nodes in a scatter plot with lasso selection (Kwon et al., 2019)

component analysis (PCA), singular value decomposition, multiple factor analysis, t-SNE, UMAP, or self-organizing maps. In our sample, PCA is the most popular technique, for example, for omics data analysis (Farg et al., 2015; Nguyen et al., 2012) or feature extraction on bacteria's infrared spectroscopy spectra (S.-Y. Ji, Jeong, et al., 2019). Reduced dimensions are typically visualized in a 2D scatter plot, which can be augmented with a density heat map, or accompanied by a parallel coordinates visualization of the original data (Farg et al., 2015; S.-Y. Ji, Jeong, et al., 2019; Figure 2d). Only Nguyen et al. (2011, 2012) use 3D scatter plots.

■ *Data mining* algorithms extract patterns from data. With sequential pattern mining, Fang et al. (2017) highlight similarities in line graphs of health sensor data, Gotz et al. (2014; Figure 2e) query event sequences and correlate them with positive and negative outcomes, Santamaría et al. (2019) highlight nucleosome patterns in sequenced chromosomes, and Klimov et al. (2015) identify risk factors for kidney damage in parallel coordinates. Furthermore, Dixit et al. (2017) optimize care pathways with process mining, and Zhao et al. (2017) mine association rules for cancer causes and visualize them in parallel coordinates.

■ *Classification* algorithms like random forest and k-nearest neighbors assign data points to a class, for example to predict rupture risk of aneurysms (Spitz et al., 2020). Their overall performance is typically visualized in a confusion matrix (e.g., Krause et al., 2018), and data points' classes are often indicated with colors (e.g., Herold et al., 2010; Nguyen et al., 2011, 2012).

The remaining five algorithmic families include specialized techniques. ■ *Artificial neural network* mainly consists of recurrent neural networks with long short-term memory to handle long-range temporal dependencies. ■ *Feature selection* contains techniques like information gain, which identify the most relevant features in a dataset. ■ *Segmentation* algorithms partition medical images into multiple segments. ■ *Anomaly detection* identifies unusual data points, for example anomalous activities in smart homes (Liao et al., 2017; Figure 2f). Finally, ■ *Other* contains algorithms like partial dependence, epidemiological models, Bayesian networks, and hidden Markov models, which do not fit in any of the previous families.

Overall, algorithmic outcomes can be visualized in many typical and alternative ways, depending on the algorithmic family and the desired insights. These insights are in turn related to the healthcare activity which is most often interpreting data, rather than predicting or monitoring.

5 | INTERACTION IN VISUAL ANALYTICS

While static visualizations may already provide interesting insights in advanced algorithms, adding interaction makes them more powerful as users can then test hypotheses, focus on particular insights, or view information from different angles. The *Interaction* group in Table 2 classifies our sample into seven interaction types proposed by Yi et al. (2007) (cfr. RQ2). *Select* (58/71) and *Abstract/elaborate* (51/71) are the most frequently supported types, whereas *Encode* (20/71) and *Explore* (19/71) the least. Most visual analytics systems support multiple interactions, and as *Select* co-occurs with all other interaction types, we present those first. Note that Lu et al. (2017) introduced *shepherding* as an interaction type, yet we discuss it separately in Section 6.

■ *Abstract/elaborate* interactions show or hide details in four ways. (1) Tooltips that pop up when hovering or clicking a visualization can show raw data (e.g., Malik et al., 2015; Xing et al., 2014) or additional visualizations (e.g., Afzal et al., 2011; Gotz et al., 2014). (2) Collapsing components removes visual clutter, for example lines in line graphs (Afzal et al., 2011) or parallel coordinates (Huang et al., 2019), identical rows in matrices (Dang et al., 2015), or similar points in scatter plots (Kwon et al., 2018). Conversely, expanding components shows extra information like individual lines instead of ribbons in parallel sets (Kwon et al., 2018), subsequences of sequential health records (Malik et al., 2015), or groups in icicle plots (Borland et al., 2020). (3) Zooming enlarges a visualization (e.g., Males et al., 2020; Kumar et al., 2015, Figure 3a). An interesting zooming variant is *lensing*, which enlarges a specific area and compresses the rest: Dang et al. (2015) use it in a large matrix of protein-biomolecule reactions. (4) To change the visualized level of a clustering hierarchy, Widanagamaachchi et al. (2017) and Behrisch et al. (2018) provide a slider and Seo and Shneiderman (2002) a bar that can be dragged along a dendrogram.

■ *Filter* interactions allow to focus on insights of interest by setting conditions on the data with check-boxes, radio buttons or sliders. Examples are: filtering network connections above a correlation threshold (Xing et al., 2014), filtering results that are statistically significant (Jönsson et al., 2019, Figure 3b), and adjusting the range of attributes in parallel coordinates by brushing axes (e.g., Yu et al., 2017) or manipulating sliders on the axes (e.g., Jönsson et al., 2019; Santamaría et al., 2008).

■ *Reconfigure* interactions change the spatial arrangement of data items in at least three ways. (1) Sorting matrices, lists and tables can reveal patterns (e.g., Dang et al., 2015) and anomalies like violations in diagnostic rules for breast cancer (Kovalerchuk et al., 2012). (2) Changing attributes. Axes in scatter plots can be configured with buttons or drop-down menus to represent different attributes (e.g., Krause et al., 2014; Abdullah et al., 2020; Stolper et al., 2014; Figure 3c). Checkboxes can change available attributes in parallel sets (e.g., Abdullah et al., 2020) or line graphs (e.g., Fang et al., 2017). (3) Repositioning data points manually can reduce occlusion (L'Yi et al., 2017; Verma et al., 2017), and automatic repositioning according to a chosen similarity metric can group similar data points (Brunker et al., 2019; S.-Y. Ji, Jeong, et al., 2019).

■ *Connect* interactions highlight associations between data items, and are often triggered by hovering. Hovering one part of a visualization can highlight other parts in the same visualization, for example connected ribbons and sets in parallel sets (e.g., Abdullah et al., 2020), neighbors in a network (Brunker et al., 2019; R. Li et al., 2020; L'Yi et al., 2017), or other features from the hovered data point (Cao et al., 2011; Gotz et al., 2011, Figure 4a). Hovering can also highlight related entities across multiple visualizations (e.g., Kakar et al., 2019; Kumar et al., 2015).

■ *Explore* interactions bring new data subsets into view. The 2D visualizations can often be panned (e.g., Afzal et al., 2011; Behrisch et al., 2018; L'Yi et al., 2017). In 3D, images can be rotated with sliders or buttons (e.g., Abbasloo et al., 2019; Jönsson et al., 2019), or the y-direction can be explored by scrolling up and down the x-z

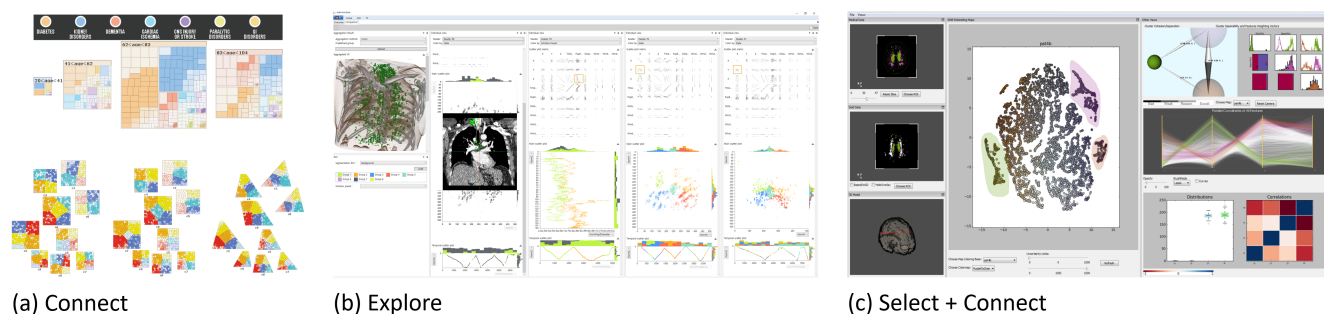


FIGURE 4 Examples of interaction with visualizations. (a) Connect a hovered cell to related cells with highlighting (Cao et al., 2011); (b) explore 3D plots with scrolling (Song et al., 2017); (c) select data in a scatter plot and update other visualizations (Raidou, Kuijf, et al., 2016)

plane (Song et al., 2017; Figure 4b). Other exploration interactions are: picking different clustering results for visualization (Kwon et al., 2018), and drilling down a Bayesian network by clicking nodes (Müller et al., 2020).

■ *Encode* interactions alter the visual representation of data, which may concern the overall visualization or the color encoding. First, some visual analytics systems switch between entirely different visualization types (e.g., Behrisch et al., 2018; Borland et al., 2020; Krause et al., 2014). Others extend a visualization, for example by adding color-coded links to a similarity scatter plot (Brunker et al., 2019), or spikes to dots in a scatter plot to represent all data dimensions (Hund et al., 2016; Figure 3d). Second, recoloring nodes in scatter plots or networks can reveal similarity to a specific node (X. Ji, Shen, et al., 2019); association strength with a particular entity (Xing et al., 2014); and cluster specifics like compactness, size, and distribution (Hund et al., 2016).

■ *Select* interactions mark data items, either manually through brushing (e.g., S. Guo et al., 2018), lasso selection (e.g., Kwon et al., 2019; Raidou, Kuijf, et al., 2016; Figure 3e), clicking on a legend (R. Guo et al., 2020), or pinning (Dingen et al., 2019; Kakar et al., 2019); or automatically based on a chosen metric (e.g., Stolper et al., 2014). Selected data are typically colored prominently to easily focus on them. For example, Herold et al. (2010) and Hinterberg et al. (2015), respectively, highlight cells on fluorescence micrographs and significant phenotype-gene expression associations that match set thresholds.

Select often precedes other interactions, hence its prominence. Geurts et al. (2015) compare the quality of several segmentation algorithms for selected segments (*Abstract/elaborate*). Liao et al. (2017) only show selected items in a radar map (*Filter*). Lamy and Tsopra (2019) reposition selected rainbow boxes, and Nguyen et al. (2011, 2012) reposition points in a scatter plot by similarity to a selected point (*Reconfigure*). Upon selecting multiple clustering results, Kwon et al. (2018) convert nodes in a scatter plot into small pie charts that reflect to which clusters the nodes belong in the different clusterings (*Encode*). Lastly, selecting data points can connect them to related data (Stolper et al., 2014; Figure 3c), or update other visualizations (*Connect*, e.g., Klemm et al., 2014; Hund et al., 2016; Raidou et al., 2015; Raidou, Kuijf, et al., 2016; Figures 3d and 4c).

To conclude, visual analytics systems support many interaction types that often co-occur. These interactions facilitate insights in algorithmic outcomes, which can in turn strengthen a user's mental model of how an advanced algorithm works.

6 | SHEPHERDING ALGORITHMS WITH VISUAL ANALYTICS

So far, we have covered two methods to gain insights in advanced algorithms: visualizing their outcomes, and interacting with the visualizations. A special interaction type is ■ *Shepherding*: guiding or controlling the algorithmic process to show algorithms' behavior under different settings (cfr. RQ3). Shepherding bridges interactive visualizations and direct explanations, because it is an example of what Mohseni et al. (2020) call "what-if explanations." Table 2 shows that less than half (32/71) of the visual analytics systems in our sample allows shepherding. This section groups those systems by their "level of integration" (Turkay et al., 2014), which indicates how seamlessly they integrate algorithms. Figure 5 shows that we found examples in the full spectrum between level two (semi-interactive) and three (tight integration). Recall that our review excluded level one systems (static visualizations or limited interaction).

Visual analytics systems of integration level two can only modify parameters or the data domain through menus or pop-up windows that obscure the visualization (e.g., Brunker et al., 2019; Santamaria et al., 2019). An illustrative example is (L'Yi et al., 2017; Figure 5a), where users can configure prediction models for miRNA-mRNA interaction in a tab completely separate from the visualization. This approach hinders swift shepherding as users constantly need to switch between configuration and visualization.

To better integrate the algorithm configuration, visual analytics systems can fix settings panels along the visualization, or use pop-ups that minimally obscure the visualization. Through radio buttons, checkboxes, text fields, or sliders, users configure algorithms and rerun them by pressing a button. Examples of modifiable aspects are: the parameter k in k -nearest neighbors clustering (Spitz et al., 2020), the number of clusters (X. Ji, Shen, et al., 2019; Figure 5b), the applied algorithm (Riegler et al., 2016), the attributes for analysis (Abbasloo et al., 2019; Dixit et al., 2017; Zhao et al., 2017), the query for pattern mining (Gotz et al., 2014; Figure 2e), and the feature weights that best distinguish ill and healthy people (Moschonas et al., 2016).

Visual analytics systems with a settings panel can integrate algorithms more tightly by rerunning them automatically after reconfiguration. For example, Clark et al. (2017) rerun statistical tests for drug dose-response analysis after altering features or the tests' sidedness, Barlowe et al. (2013) rerank histograms after modifying the number of bins,

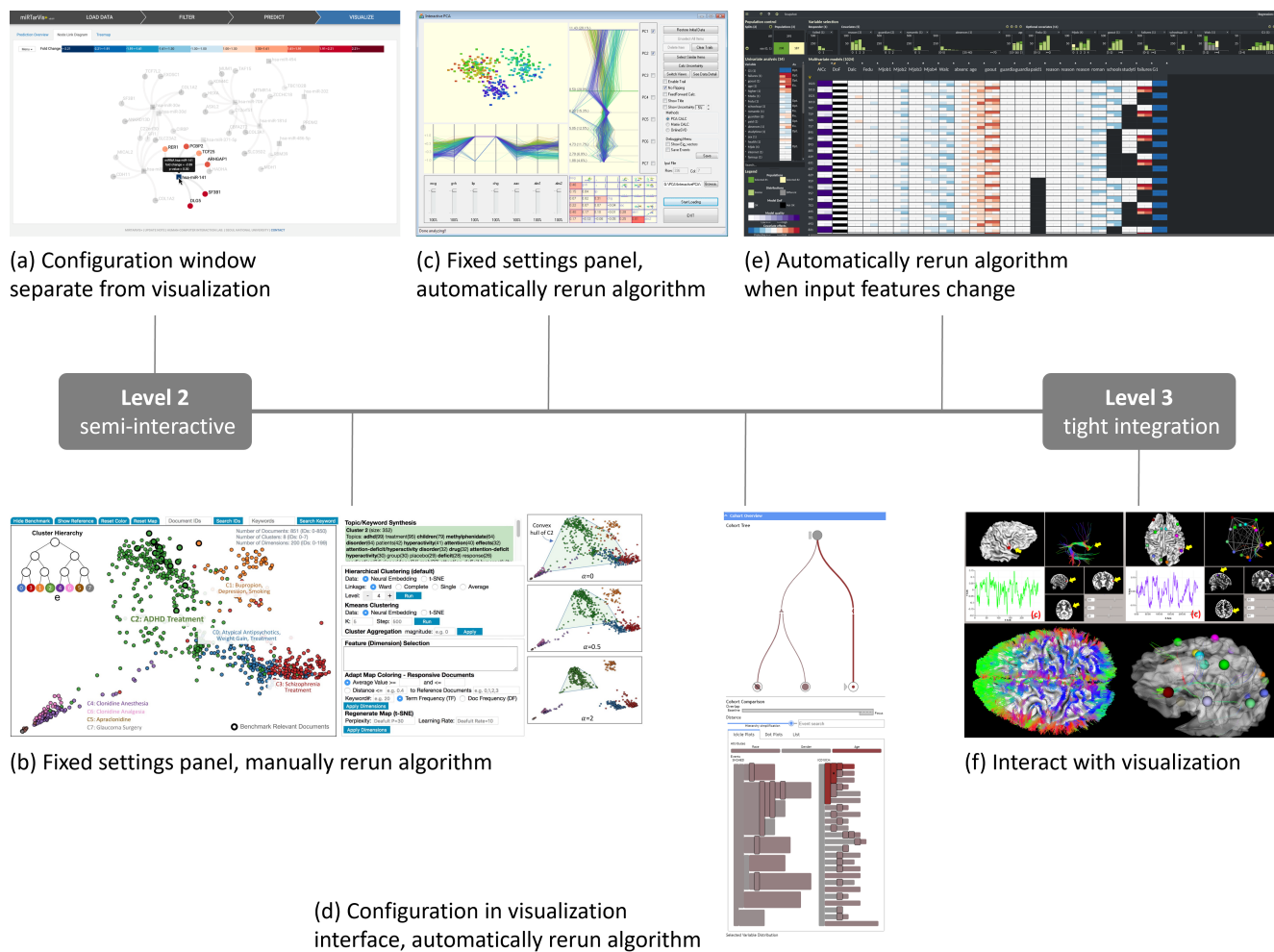


FIGURE 5 Examples of shepherding, ordered by the level of integration in the visual analytics system (Turkay et al., 2014). Semi-interactive examples are situated left; the more to the right, the tighter the integration. (a) L'Yi et al. (2017), (b) X. Ji, Shen, et al. (2019), (c) Jeong et al. (2009), (d) Borland et al. (2020), (e) Dingen et al. (2019), (f) K. Li et al. (2012)

Feller et al. (2018) reapply clustering after changing the number of clusters, Jeong et al. (2009) and S.-Y. Ji et al. (2017; Figure 5c), change the contribution of variables in weighted PCA, and S. Guo et al. (2018) adjust the clustering level of medical event sequences to find meaningful groupings and to understand their sensitivity.

To further integrate algorithms into visual analytics systems, visualizations can themselves incorporate configuration functionalities like sliders to adjust how aggressively sequential events are grouped (Gotz et al., 2020; Borland et al., 2020; Figure 5d), drop-down menus to configure dimension reduction techniques (Abdullah et al., 2020; Kwon et al., 2018), and textfields to set the maximal cohort size for cohort clustering (C.-W. Huang et al., 2015; M. Huang et al., 2019).

Four visual analytics systems approach the highest integration level: they automatically update algorithmic outcomes when input changes. First, R. Li et al. (2020) show predicted risk of heart failure in a line chart, and add a line for the updated risk after removing or adding drugs. (Kwon et al., 2019) is similar, though its predictions need to be rerun manually. Next, Afzal et al. (2011) compare diseases' mortality and infection rates under different epidemiological parameters and measures. Last, Dingen et al. (2019; Figure 5e), allow to drag variables into a dedicated panel to automatically generate logistic regression models, and compare those models across groups.

Finally, visual analytics systems of the third integration level shepherd algorithms through direct interaction with visualizations. First, K. Li et al. (2012; Figure 5f), allow to relocate and edit regions of interest in a 3D image of the brain, after which the strength of connections between them is recomputed. Second, Hund et al. (2016; Figure 3d), project clustering results under different distance measures, and rerun the algorithms when users filter the data. Third,

TABLE 3 Classification of 10 visual analytics systems with direct explanations in our sample according to explanation type, explanation scale, explainer, and target user

	TYPE	SCALE	EXPLANATOR	USER
	Model-agnostic Model-specific	Global Local Model inspection	Activation maximization Contextual decomposition Decision rules Decision tree Feature importance Partial dependence plot Sensitivity analysis Visualization	AI novices Data experts AI experts
Alsaad et al. (2019)	•	•	•	• •
Hur et al. (2020)	•	•	•	• •
Krause et al. (2014)	•	•	•	• •
Krause et al. (2016)	•	• •	•	• •
Krause et al. (2018)	•	• •	•	• •
Kwon et al. (2018)	•	•	•	• •
Lamy and Tsopra (2019)	•	•	•	• •
Müller et al. (2020)	•	• •	•	• •
Nauta et al. (2020)	•	•	•	• •
Von Landesberger et al. (2013)	•	•	•	•
Total	1 9	4 6 3	1 1 1 1 4 1 1 2	1 8 10

Gotz et al. (2011) and Cao et al. (2011; Figure 4a) provide rich interactions to refine clusters of patients: users can filter features, merge clusters with lasso selection, drag patients out of clusters, and automatically remove patients far from the cluster center.

7 | DIRECTLY EXPLAINING ALGORITHMS WITH VISUAL ANALYTICS

Besides visualizing and interacting with algorithmic outcomes, a final technique to better understand advanced algorithms is to directly explain how they work (cfr. RQ4). Our sample includes 10 visual analytics systems with explicit explanations: Table 3 classifies them by type, scale, explainer, and target user.

7.1 | Type

All but one explanation techniques are model-specific. Some of them can be applied to other algorithms of the same family (e.g., all feature selection algorithms), but only *Prospector* (Krause et al., 2016; Figure 6a) is fully model-agnostic. *Prospector* shows how a prediction is affected when feature values are perturbed with color-coded sliders that correspond to partial dependence plots.

7.2 | Scale

Zooming in on explanations' scale, our sample mainly contains local explanations (6/10). Three systems explain artificial neural networks (ANNs) on a different scale. (1) Nauta et al. (2020) globally explain how an ANN predicts coma outcome: for a fixed epoch and a fixed hidden layer, all input activations are projected onto a scatter plot, and users can then select clusters to train a decision tree that distinguishes them. (2) Alsaad et al. (2019) use contextual decomposition to locally explain how a long short-term memory (LSTM) network predicts asthma based on clinical visits: each visit's contribution to the prediction is visualized in a heat map matrix that also highlights the most predictive subset of visits. (3) Hur et al. (2020) apply model inspection to explain a LSTM that predicts heart failure or heart surgery based on medical pathways: for the average or a specific patient, an attention heat map shows the variable weights in all time steps of the LSTM.

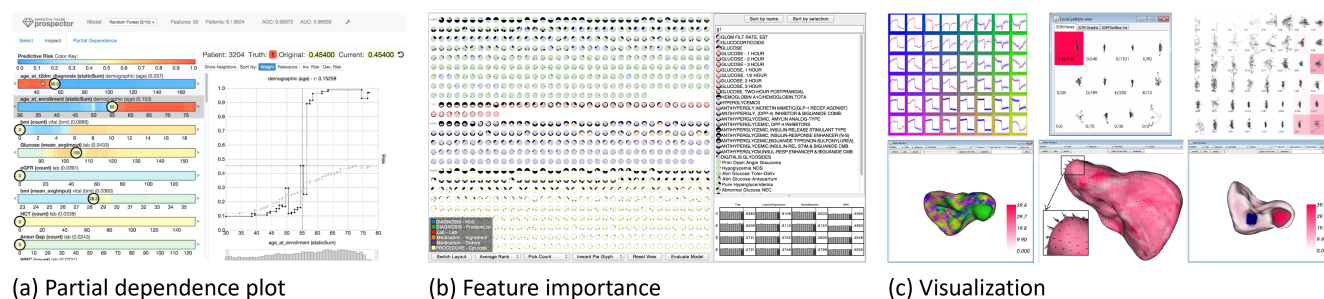


FIGURE 6 Examples of direct explanations in visual analytics systems. (a) Sliders show how perturbed inputs impact the prediction (Krause et al., 2016); (b) circular glyphs with quadrants of feature importance scores (Krause et al., 2014); (c) segmented mesh color-coded by similar landmark movements (Von Landesberger et al., 2013)

7.3 | Explainer

Four visual analytics systems in our sample use feature importance for explanations: they assign scores to features to indicate how strongly they impact the algorithmic outcome. (a) Müller et al. (2020) predict suitable cancer treatments with a Bayesian network, and determine the global and local relevance of evidence with sensitivity analysis. Users can also see the impact of updating or adding evidence in donut charts. (b) To select features that best predict diabetes, Krause et al. (2014; Figure 6b) score feature importance with information gain, Fisher score, odds ratio, and relative risk. These four scores are computed in 10 cross-validation folds, and visualized as quadrants of a circular bar chart. (c) Krause et al. (2018) predict hospital admission with binary classifiers, and explain them with decision rules, which consist of feature sets that change the prediction outcome when removed. Users can inspect these decision rules in a matrix of data items (rows) and features (columns), ordered by Gini feature importance. (d) After clustering, Kwon et al. (2018) apply ANOVA to identify significant relationships between features and clusters, and use the F -values as importance values to rank features.

7.4 | User

Most of the collected explanation methods (8/10) are designed for both data experts and AI experts. Von Landesberger et al. (2013; Figure 6c), specifically target AI experts, who can detect drawbacks of segmentation algorithms through visualizations of landmark movements. Lamy and Tsopra (2019) also explain through visualization, which seems the only approach suitable for AI novices: they visualize ANNs without hidden layers as rainbow boxes. These boxes symbolize connections in the ANN, and their height equals the weight of those connections.

8 | OBSERVATIONS, OPPORTUNITIES, AND CHALLENGES

Our review investigated four methods to obtain insights in advanced algorithms: (a) visualizing their outcomes and (b) interacting with these visualizations, (c) shepherding them, and (d) directly explaining them. These methods are not always clearly separable as shepherding can be a kind of interaction with visualizations, and visualizations can also act as direct explanations. Put differently, there is a fine line between getting insights in algorithmic outcomes and in the inner logic of algorithms themselves. This section answers our research questions, and positions them in the broader, multidisciplinary context of Figure 7.

8.1 | Visualizing outcomes: Many algorithm-dependent possibilities

Section 4 showed that algorithmic outcomes can be visualized in many ways (RQ1), ranging from basic scatter plots to original custom glyphs. Some visualizations often occur together with specific algorithms, for example dendrograms with hierarchical clustering, and projection plots with dimension reduction. This co-occurrence seems to hold in general: visualization approaches strongly depend on the algorithmic family and the healthcare activity. We observed that

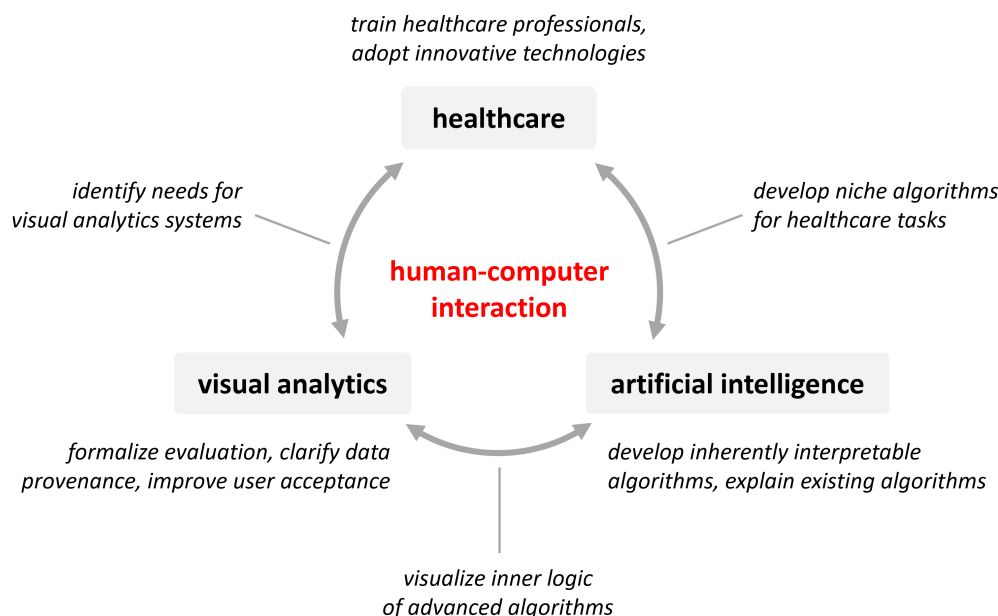


FIGURE 7 Opportunities and challenges for visual analytics, artificial intelligence, and healthcare. Human-computer interaction can mediate the dialogue between these three communities

visual analytics systems for a healthcare context mainly rely on mainstream interpretative algorithms such as clustering, dimension reduction and classical statistics, resulting in few visual analytics systems for predictive activities, and even none for monitoring chronic diseases like diabetes. On the one hand, this suggests room for adopting more specialized state-of-the-art AI techniques like transformer neural networks, Bayesian networks, and natural language processing. On the other hand, the absence of monitoring systems in our sample suggests a gap for further research. This gap is related to a second observation: most of the visual analytics systems in our sample target healthcare professionals, but as technology increasingly facilitates self-monitoring, laypeople are likely to become an important target group too. Future studies could thus investigate how visual analytics may fit the needs of laypeople.

8.2 | Interacting with visualizations: Sufficient or too much?

Section 5 demonstrated that existing visual analytics systems incorporate all seven interaction types from Yi et al. (2007) (RQ2): *Abstract/elaborate*, *Connect*, *Encode*, *Explore*, *Filter*, *Reconfigure*, and *Select*. The frequent appearance of *Abstract/elaborate* presumably originates from the widespread details-on-demand mantra in information visualization (Shneiderman, 2003). All interaction types exist in different forms, and are often combined in highly interactive visualizations. While this is encouraged to facilitate insights in algorithmic outcomes, some developers of visual analytics systems in our sample (e.g., Kwon et al., 2019) intensively collaborated with healthcare professionals and note that caution is needed: end-users are not always looking for highly exploratory, information-heavy interfaces that are interesting from a visualization perspective, but are too complex for their needs. Instead, some healthcare contexts may require visual analytics systems that act as fellow healthcare experts and point out interesting cases in the data. Thus, tailoring the amount of interaction in visual analytics systems is part of the broader challenge to involve end-users in every stage of the design process and identify their needs. This may improve user acceptance and enhance trust in the proposed system (Abdullah et al., 2020).

8.3 | Shepherd algorithms: A higher-order interaction

Section 6 showed that algorithms can be shepherded by tuning algorithmic parameters or modifying input (RQ3). Visual analytics systems can integrate shepherding along a continuum that ranges from separating shepherding and visualizations to tightly connecting both. Despite its potential to provide what-if explanations, optimize existing models, or build new meaningful models, shepherding was relatively uncommon in our sample. This could be due to a trade-off that rises when

the amount of transparency and shepherding freedom is determined: giving too much control to non-trained users can overwhelm them or incur misleading outcomes resulting from overfit models. Blindly playing around with parameters may be harmless for exploratory contexts such as clustering similar documents (X. Ji, Shen, et al., 2019) or medical images (Riegler et al., 2016), but can have severe consequences in delicate predictive contexts. Therefore, AI experts should inform healthcare professionals about the applicability, strengths and weaknesses of AI models, and visual analytics developers should help in training healthcare professionals to use their systems. User-studies in our sample endorse the need for such training (e.g., Cao et al., 2011; Gotz et al., 2020; Kwon et al., 2021). Another possibility is to develop different versions of visual analytics systems and only provide shepherding functionalities in specific contexts.

8.4 | Direct explanations: Rare yet promising

Section 7 revealed two interesting observations about direct explanations (RQ4). First, few examples were present in our sample, yet all of them involved visualizations. Second, AI novices are so far often neglected. Although directly explaining advanced algorithms to AI novices seems challenging, future research could for example investigate whether conversational design and example-based explanations give solace (Ribera & Lapedriza, 2019). In that way, patients may better understand personalized health plans proposed by their clinician, potentially by comparing their health measurements against similar patients. Of course, AI experts remain an important target group as well: direct explanations in interactive visual analytics systems can help them explain existing black-box models, obtain information about the inner logic of advanced algorithms, and design algorithms that are better interpretable. Regarding the latter, collaborating with healthcare professionals is essential to learn what interpretability means in their context.

9 | CONCLUSION

XAI is extremely relevant in our current age of algorithms and massive data collection, and visual analytics has proven to play an important role in the quest for explainability. Healthcare has acknowledged the value of visual analytics in many applications, but may not have taken enough advantage of this exciting field yet (West et al., 2015). For example, Bonnett et al. (2019) showed that risk prediction is still dominated by simple static visualizations like point score systems and nomograms. In addition, our review suggests a lack of interactive visual analytics systems for monitoring, and systems that target laypeople.

Our review also reveals that visual analytics holds many opportunities for XAI in healthcare by providing insights in advanced algorithms through visualization, interaction, shepherding, and direct explanations. Yet, complex challenges remain: many healthcare stakeholders are involved in the visual analytics process (Kolyshkina & Simoff, 2021), domain practices should be respected, domain expertise is often required to correctly interpret algorithmic results, and explanation techniques should be tailored to the application domain and target users. These XAI challenges cannot be solved in isolation, so we encourage the visual analytics, AI and healthcare communities to further reach out to each other, and we invite the human-computer interaction community to help mediating this fascinating interdisciplinary dialogue.

ACKNOWLEDGMENT

This work was supported by the Research Foundation-Flanders (FWO) grant G0A3319N and the Slovenian Research Agency grant ARRS-N2-0101. We thank all researchers who provided screenshots of their visual analytics systems and gave permission to reproduce them here in Figures 1–6.

CONFLICT OF INTEREST

The authors have declared no conflicts of interest for this article.

AUTHOR CONTRIBUTIONS

Jeroen Ooge: Data curation (lead); methodology (lead); writing - original draft (lead). **Gregor Stiglic:** Writing-review & editing (supporting). **Katrien Verbert:** Supervision (equal); writing-review & editing (supporting).

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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How to cite this article: Ooge, J., Stiglic, G., & Verbert, K. (2022). Explaining artificial intelligence with visual analytics in healthcare. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 12(1), e1427. <https://doi.org/10.1002/widm.1427>